The stochastic Taylor tree in every positive dimension: coefficient continuity and ordered remainders (Programme S, tranche A1–A2)

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Abstract

Units 270–277 lift the stochastic clause of `thm:strataempiricalexpansion` from the single-chart $d = 2$ statements of the normal-block programme to arbitrary normal dimension, on top of the completed Taylor tree. On the Banach space $E_b \times E_b$ of weighted-ℓ^1 phase and amplitude data ($\sum_\gamma |c_\gamma| b^{|\gamma|} < \infty$, modelled as ℓ^1 over a disjoint union of multi-index sets in rescaled coordinates), every canonical coefficient $C_{\mu,j}$ of $N^{-\mu}(\log N)^j$ in the expansion of the original box integral is Lipschitz on balls, hence continuous and Borel measurable (Headline XXXIV, a weighted-ℓ^1 substitute for the paper's cited-but-unstated `prop:convergence`). Ordering the terms by $(\nu, q) \prec (\mu, j)$ iff $\nu < \mu$ or ($\nu = \mu$, $q > j$), the ordered normalised remainders converge to $C_{\mu,j}$ uniformly on data balls, and for measurable random data $X_n \Rightarrow Z$ in $E_b \times E_b$ and sample sizes $N_n \to \infty$, finite coefficient vectors and ordered normalised remainders converge in distribution, jointly for finitely many targets (Headline XXXV). Reviews v27 and v28: pass. Eight units against a cap of eighteen. No `sorry`, no additional `axiom`; 3405 jobs; pin `caa3727`.

1 The design (Astra #33)

After the Taylor tree was frozen, the instruction to continue formalising the paper pointed at §4.3. Astra #33 ranked the stochastic Taylor tree first: (A1) continuity of the coefficient functionals, (A2) convergence in distribution of coefficients and ordered remainders, with a hard review after the third unit; then (A3) tangential integration and chart assembly and (A4) posterior division behind statement gates. Two design points shaped everything: use a genuine Banach space with the frozen `CoeffFamily` theorems as its analytic interface, and treat the constant phase $a = \xi(0)$ as a perturbation in its own right — the Stage 4 stability gate had held it fixed.

2 A1: the data space and the Lipschitz estimate (units 270–272)

Data space (u270). `DataSpace d` is real ℓ^1 over $(\mathrm{Fin}\, d \to \mathbb{N}) \sqcup (\mathrm{Fin}\, d \to \mathbb{N})$. Its raw coordinates are exactly the *rescaled unit-box families* $\mathrm{scale}\, c_\xi\, b$, $\mathrm{scale}\, c_\eta\, b$ of the Taylor tree, so the box families are recovered by dividing by $b^{|\gamma|}$ and the norm is the sum of the two masses. The canonical map `dataBoxCoeff` is `boxCoeff` of the box families; through `boxSpectralSum_eq` it is literally the coefficient of $N^{-\mu}(\log N)^j$ for the original integral, including the binomial conversion of $\log(N b^{2|k|})$.

The gate (u271). With $f = c_\eta * J^{*p}$, J the constant-free phase, the coefficient series term is $\beta^p/p!\, T_p(a; f)$. The difference of two data splits as $[T_p(a'; f') - T_p(a; f')] + T_p(a; f' - f)$. The first piece is new: by the mean value theorem on $[-R, R]$ and $\partial_a\, \mathrm{fluctMoment} = \beta\, \mathrm{fluctMoment}(p{+}1)$, $|\mathrm{fluctMoment}(a) - \mathrm{fluctMoment}(a')| \leq \beta M_{\mu,n,p+1}(R)|a - a'|$, which propagates through the kernel

S_p and the functional T_p. The second piece uses linearity and the mass algebra of the Cauchy product, $\mathrm{mass}(f'-f) \le \Delta_\eta R^p + R\,pR^{p-1}\Delta_\xi$. The three series in p are summed in closed form by the Tonelli identity $\sum_p (\beta R)^p/p!\, M_{\nu,n,p}(R) = M_{\nu,n,0}(2R)$ and its shifted forms, giving the ballwise constant $K_k D\big(2\beta R M_{\mu+1/2,n,0}(2R) + M_{\mu,n,0}(2R)\big)$.

Headline XXXIV (u272). Transport to the data space (masses $\le \|x\|$, differences $\le \|y-x\|$) and the finite log conversion give Lipschitz continuity on every closed ball, continuity (the closed ball of radius $\|x\|+1$ is a neighbourhood), Borel measurability, and continuity of finite coefficient vectors.

3 A2: the deterministic core and the probability (units 273–277)

Uniform constants (u273–274). The cutoff constant is monotone in $|\xi(0)|$ and the two masses, so on a data ball it is bounded by $\mathrm{cutoffBound}(R, R, R)$; the box integral itself is Lipschitz in the data for fixed N (the integrand is continuous on the closed cube by the Weierstrass M-test, integrable on the box, and $|e^{A+s\xi} - e^{A+s\xi'}| \le e^{\beta\sqrt{T}R}\beta\sqrt{T}|\xi-\xi'|$ with $T = Nb^{2|k|}$), hence measurable.

Ordered remainders (u275). With the cutoff $L = \mu+1$ the retained index set is the finite $\Lambda_{\mu+1} \times \{0, \ldots, n\}$; the predecessors of (μ, j) are its $\prec$-filter, independent of the cutoff (u277). $Z - \sum_{\prec} -C_{\mu,j}N^{-\mu}(\log N)^j$ is the cutoff error plus the retained terms with $\nu > \mu$ or ($\nu = \mu$, $q < j$); after normalisation these are $O(N^{-(\nu-\mu)}(1+\log N)^n)$ and $O(1/\log N)$ with constants uniform on the ball, and the cutoff error is $O(N^{-1}(1+\log N)^n)$. The explicit majorant is exported (`abs_orderedRemainder_sub_le`) and gives `TendstoUniformlyOn` on the closed ball.

Headline XXXV (u276–277). Coefficient vectors converge in distribution by Mathlib's continuous mapping theorem. For the remainders, the norms $\|X_n\|$ are bounded in probability (portmanteau on the closed sets $\{\|\cdot\| \ge m\}$, a generic copy of the $d=2$ argument), so uniform convergence on balls gives convergence in probability of $R_{N_n}(X_n) - C(X_n)$ to zero, and `tendstoInDistribution_of_tendstoInMeasure_sub` (Slutsky) finishes. The joint version for finitely many targets uses the sup norm on $\mathrm{Fin}\, m \to \mathbb{R}$ and `Filter.eventually_all`.

4 Reviews and non-claims

Review v27 (u270–272): pass, pass, qualified pass; it asked for the canonical cutoff to be restated with `dataBoxCoeff`, for measurability of the integral and remainders, for the finite predecessor set and descending-log ordering, for uniform convergence on balls with thresholds, and for the boundedness-in-probability and Slutsky steps — all done in u273–277. Review v28 (u273–276): pass on all four units, "no mathematical blocker". Non-claims: the result is conditional on convergence in distribution of the data in the weighted-ℓ^1 topology at the fixed box radius b (nothing is derived from Hypothesis I; no common analytic radius is manufactured for random phases); one chart, no tangential integration or assembly; no Gaussianity of the limit; deterministic sample sizes $N_n \ge 0$ in the paper's convention; the normalised quotient is meaningful above the thresholds $N \ge e$, $Nb^{2|k|} \ge 1$.

5 Lessons

- **Use the induction structure you already have.** Every estimate here is a transport of a frozen Taylor-tree theorem through the identification "raw ℓ^1 coordinates = rescaled unit-box families"; no new analysis of the integral was needed except the mean-value step in $\xi(0)$.
- **Mean value beats differentiation under the integral.** The phase perturbation is a one-line consequence of the existing derivative theorem for `fluctMoment`.
- **Deterministic first, probability last.** Stating the uniform-on-balls theorem separately made the probability step a two-lemma composition and let the reviewer check the analysis on its own.
- **Tooling.** In this repository `lake env lean` cannot see the artifact-cached package oleans; build single modules with `lake build Grammar.X`. Long-line lint must gate the commit (three fix-up commits were needed before the gate was made mechanical). GPT consults must run in the foreground on this host.
- **Mathlib's probability API is enough.** `TendstoInDistribution` with its continuous mapping, Slutsky and portmanteau lemmas covered A2 without new infrastructure.